ECEN-635: Homework 1

Due on September 12, 2025 at 4:30 pm

Instructor: Dr. Robert Nevels

Brody Jordan
431005357

Problem 1

Convert the following time domain expressions for the electric and magnetic field intro frequency and domain expressions. Show each mathematical step in your calculations. A miraculous answer without the complete calculation will be judged harshly.

(a) $\mathbf{E}(t) = 4\cos(\omega t - 28z)\hat{\mathbf{y}}$ V/m
(b) $\mathbf{E}(t) = 9\sin(\omega t - 23z)\hat{\mathbf{x}}$ V/m
(c) $\mathbf{H}(t) = 7\cos(\omega t - 23z)\hat{\mathbf{y}}$ mA/m *changed polarization from x to y*
(d) $\mathbf{E}(t) = 4.6e^{-48\pi z}\cos(\omega t - 139z + \pi/4)\hat{\mathbf{y}}$ V/m

Solution

Part (a)

$$\begin{aligned}\cos(\omega t - 28z) &= \Re\{e^{j(\omega t - 28z)}\} \\ &= \Re\{e^{j\omega t}e^{-j28z}\} \\ &\Rightarrow \quad \mathbf{E}(\omega) = 4e^{-j28z}\hat{\mathbf{y}} \ \text{V/m}\end{aligned}$$

Part (b)

$$\begin{aligned}9\sin(\omega t - 23z) &= -9\cos(\omega t - 23z + \pi/2) \\ &= \Re\{-9e^{j(\omega t - 23z + \pi/2)}\} \\ &\Rightarrow \quad \mathbf{E}(\omega) = -9e^{j\pi/2}e^{-j23z}\hat{\mathbf{x}} \ \text{V/m}\end{aligned}$$

Part (c)

$$\begin{aligned}7\cos(\omega t - 23z) &= \Re\{7e^{j(\omega t - 23z)}\} \\ &= \Re\{7e^{j\omega t}e^{-j23z}\} \\ &\Rightarrow \quad \mathbf{H}(\omega) = 7e^{-j23z}\hat{\mathbf{y}} \ \text{mA/m}\end{aligned}$$

Part (d)

$$\begin{aligned}4.6e^{-48\pi z}\cos(\omega t - 139z + \pi/4) &= 4.6e^{-48\pi z}\Re\{e^{j(\omega t - 139z + \pi/4)}\} \\ &= 4.6e^{-48\pi z}\Re\{e^{j\omega t}e^{-j139z}e^{j\pi/4}\} \\ &\Rightarrow \quad \mathbf{E}(\omega) = 4.6e^{-(48\pi + j139)z}e^{j\pi/4}\hat{\mathbf{y}} \ \text{V/m}\end{aligned}$$

Problem 2

Convert the following frequency domain expressions to the time domain. Show each mathematical step in your calculation.

(a) $\mathbf{E} = 15e^{-j23z}\hat{\mathbf{x}}$ V/m
(b) $\mathbf{E} = 32e^{-(52+j135)z}\hat{\mathbf{y}}$ V/m
(c) $\mathbf{H} = [(2+j2)\hat{\mathbf{x}} + (1-j3)\hat{\mathbf{y}}]e^{-j245z}$ A/m
(d) $\rho = 3je^{j\pi/2}$ C/m^3
Be sure to place each of the vector components in simplest magnitude and phase form before converting to the time domain.

Solution

Part (a)

$$\begin{aligned}\mathbf{E} &= 15e^{-j23z}\hat{\mathbf{x}} \text{ V/m}\\ &= \Re\{15e^{j(\omega t-23z)}\}\hat{\mathbf{x}} \text{ V/m}\\ &\Rightarrow \quad \mathbf{E}(t) = 15\cos(\omega t - 23z)\hat{\mathbf{x}} \text{ V/m}\end{aligned}$$

Part (b)

$$\begin{aligned}\mathbf{E} &= 32e^{-(52+j135)z}\hat{\mathbf{y}} \text{ V/m}\\ &= 32e^{-52z}e^{-j135z}\hat{\mathbf{y}} \text{ V/m}\\ &= \Re\{32e^{-52z}e^{j(\omega t-135z)}\}\hat{\mathbf{y}} \text{ V/m}\\ &\Rightarrow \quad \mathbf{E}(t) = 32e^{-52z}\cos(\omega t - 135z)\hat{\mathbf{y}} \text{ V/m}\end{aligned}$$

Part (c)

$$\begin{aligned}\mathbf{H} &= [(2+j2)\hat{\mathbf{x}} + (1-j3)\hat{\mathbf{y}}]e^{-j245z} \text{ A/m}\\ &= [2\sqrt{2}e^{j\pi/4}\hat{\mathbf{x}} + \sqrt{10}e^{-j\arctan(3)}\hat{\mathbf{y}}]e^{-j245z} \text{ A/m}\\ &= \Re\{2\sqrt{2}e^{j(\omega t-245z+\pi/4)}\hat{\mathbf{x}} + \sqrt{10}e^{j(\omega t-245z-\arctan(3))}\hat{\mathbf{y}}\} \text{ A/m}\\ &\Rightarrow \quad \mathbf{H}(t) = 2\sqrt{2}\cos(\omega t - 245z + \pi/4)\hat{\mathbf{x}} + \sqrt{10}\cos(\omega t - 245z - \arctan(3))\hat{\mathbf{y}} \text{ A/m}\end{aligned}$$

Part (d)

$$\begin{aligned}\rho &= 3je^{j\pi/2} \text{ C/m}^3\\ &= \Re\{3e^{j(\omega t+\pi/2+\pi/2)}\} \text{ C/m}^3 \qquad \text{since } j = e^{j\pi/2}\\ &\Rightarrow \quad \rho(t) = 3\cos(\omega t + \pi) \text{ C/m}^3\end{aligned}$$

Problem 3

Assuming a homogeneous region where there is no current, charge, or loss:

(a) Use Maxwell's equations to convert the frequency domain electric field expression in 2(b) to the corresponding magnetic field. Show all mathematical steps.

(b) Use Maxwell's equations to convert the frequency domain magnetic field in 2(c) into a frequency domain electric field into a frequency domain magnetic field. Show all mathematical steps.

Solution

Part (a)

We have Maxwell's equations which provides a relationship between the electric and magnetic fields, and we have an electric field from 2(b):

$$\begin{cases} \nabla \times \mathbf{E} = -j\omega \mathbf{H} \\ \mathbf{E} = 32e^{-(52+j135)z}\hat{\mathbf{y}} \end{cases}$$

Since this electric field has only a y-component, the curl becomes

$$\nabla \times \mathbf{E} = \left(-\frac{\partial}{\partial z}E_y\right)\hat{\mathbf{x}} + \left(\frac{\partial}{\partial x}E_y\right)\hat{\mathbf{z}}$$

The partial derivative of E_y with respect to x is zero, so our curl simplifies to

$$\begin{aligned} \nabla \times \mathbf{E} &= -\frac{\partial}{\partial z}\left(32e^{-(52+j135)z}\right)\hat{\mathbf{x}} + 0\hat{\mathbf{z}} \\ &= -32(-(52+j135))e^{-(52+j135)z}\hat{\mathbf{x}} \\ &= 32(52+j135)e^{-(52+j135)z}\hat{\mathbf{x}} \end{aligned}$$

Now, substituting this into Maxwell's equation, we can find the magnetic field

$$\begin{aligned} 32(52+j135)e^{-(52+j135)z}\hat{\mathbf{x}} &= -j\omega \mathbf{H} \\ \implies \mathbf{H} &= -\frac{32(52+j135)}{j\omega}e^{-(52+j135)z}\hat{\mathbf{x}} \end{aligned}$$

Part (b) continued on next page...

Part (b)

Now we have a magnetic field from 2(c) and Maxwell's equations:

$$\begin{cases} \nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E} \\ \mathbf{H} = [(2+j2)\hat{\mathbf{x}} + (1-j3)\hat{\mathbf{y}}]e^{-j245z} \end{cases}$$

Calculating the curl of $\mathbf{H}$, we have

$$\begin{aligned}
\nabla \times \mathbf{H} &= \left(-\frac{\partial}{\partial z}H_y\right)\hat{\mathbf{x}} + \left(\frac{\partial}{\partial z}H_x\right)\hat{\mathbf{y}} + \left(\frac{\partial}{\partial x}H_y - \frac{\partial}{\partial y}H_x\right)\hat{\mathbf{z}} \\
&= \left(-\frac{\partial}{\partial z}H_y\right)\hat{\mathbf{x}} + \left(\frac{\partial}{\partial z}H_x\right)\hat{\mathbf{y}} \\
&= \left(-\frac{\partial}{\partial z}\left((1-j3)e^{-j245z}\right)\right)\hat{\mathbf{x}} + \left(\frac{\partial}{\partial z}\left((2+j2)e^{-j245z}\right)\right)\hat{\mathbf{y}} \\
&= \left(-(1-j3)(-j245)e^{-j245z}\right)\hat{\mathbf{x}} + \left((2+j2)(-j245)e^{-j245z}\right)\hat{\mathbf{y}} \\
&= (735+j245)\, e^{-j245z}\hat{\mathbf{x}} + (490-j490)\, e^{-j245z}\hat{\mathbf{y}}
\end{aligned}$$

Now relating it to the electric field using Maxwell's equations

$$\begin{aligned}
&(735+j245)\, e^{-j245z}\hat{\mathbf{x}} + (490-j490)\, e^{-j245z}\hat{\mathbf{y}} = j\omega\epsilon\mathbf{E} \\
\implies &\mathbf{E} = \frac{1}{j\omega\epsilon}\left[(735+j245)\, e^{-j245z}\hat{\mathbf{x}} + (490-j490)\, e^{-j245z}\hat{\mathbf{y}}\right]
\end{aligned}$$

Problem 4

Use 1(b) and the corresponding electric field found in problem 1(c) to calculate the time average power density. Show all mathematical steps.

Solution

We have the electric and magnetic fields from problems 1(b) and 1(c) and a relationship between them:

$$\begin{cases} \mathbf{E}(t) = 9\sin(\omega t - 23z)\hat{\mathbf{x}} \ \ \text{V/m} & \text{from 1(b)} \\ \mathbf{H}(t) = 7\cos(\omega t - 23z)\hat{\mathbf{y}} \ \ \text{mA/m} & \text{from 1(c)} \\ \mathscr{S}_{av} = \mathbf{S} = \frac{1}{2}\Re\{\mathbf{E} \times \mathbf{H}^*\} \end{cases}$$

In the frequency domain and in base SI units, we have

$$\begin{cases} \mathbf{E}(\omega) = 9e^{-j\pi/2}e^{-j23z}\hat{\mathbf{x}} & \text{from 1(b)} \\ \mathbf{H}(\omega) = 7 \cdot 10^{-3}e^{-j23z}\hat{\mathbf{y}} & \text{from 1(c)} \end{cases}$$

Now we can compute $\mathbf{E} \times \mathbf{H}^*$. Their cross will only have a z-component since $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$.

$$\begin{aligned} \mathbf{E} \times \mathbf{H}^* &= (9e^{-j\pi/2}e^{-j23z}\hat{\mathbf{x}}) \times (7 \cdot 10^{-3}e^{j23z}\hat{\mathbf{y}}) \\ &= 63 \cdot 10^{-3}e^{-j\pi/2}(\hat{\mathbf{x}} \times \hat{\mathbf{y}}) \\ &= 63 \cdot 10^{-3}e^{-j\pi/2}\hat{\mathbf{z}} \\ &= -j63 \cdot 10^{-3}\hat{\mathbf{z}} \end{aligned}$$

So the time average power density is

$$\begin{aligned} \mathbf{S} = \frac{1}{2}\Re\{\mathbf{E} \times \mathbf{H}^*\} &= \frac{1}{2}\Re\{-j63 \cdot 10^{-3}\hat{\mathbf{z}}\} \\ &= \frac{1}{2}(0)\hat{\mathbf{z}} \\ &= 0\hat{\mathbf{z}} \ \ \text{W/m}^2 \end{aligned}$$

Problem 5

Starting with the frequency domain form of Maxwell's equations, derive the magnetic field vector Helmholtz equation. Number your Maxwell equations and at every step write briefly which Maxwell equation and/or which identity you are using from the Vectory Identities section of Appendix II.3.1 (pg. 958). Assume there is no electric or magnetic current and no electric or magnetic charge.

Solution

Here are Maxwell's equations with no charges

$$\begin{cases} \nabla \times \mathbf{E} = -j\omega\mu\mathbf{H} & (1) \\ \nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E} & (2) \\ \nabla \cdot \mathbf{E} = 0 & (3) \\ \nabla \cdot \mathbf{H} = 0 & (4) \end{cases}$$

Start by taking the curl of equation (2)

$$\nabla \times (\nabla \times \mathbf{H}) = \nabla \times (j\omega\epsilon\mathbf{E}) \tag{5}$$

Using vector identity (Appendix II-51) on the left hand side of (5)

$$\nabla \times (\nabla \times \mathbf{H}) = \nabla(\nabla \cdot \mathbf{H}) - \nabla^2\mathbf{H} \tag{6}$$

We substitute (6) into (5) to get

$$\nabla(\nabla \cdot \mathbf{H}) - \nabla^2\mathbf{H} = \nabla \times (j\omega\epsilon\mathbf{E}) \tag{7}$$

Now substituting equation (1) and (4) into (7)

$$-\nabla^2\mathbf{H} = j\omega\epsilon(-j\omega\mu\mathbf{H}) \tag{8}$$

Simplifying (8) by grouping the constants into k^2 gives us

$$\nabla^2\mathbf{H} + \omega^2\mu\epsilon\mathbf{H} = 0 \qquad \Rightarrow \qquad \nabla^2\mathbf{H} + k^2\mathbf{H} = 0 \tag{9}$$

Which is the magnetic field vector Helmholtz equation. :)